

Ryan Yan Yan 闫闫 (she/her)

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I study how reward processing and decision-making interact with subjective experience of affect, both as they fluctuate within individuals over time and in the context of affective psychopathology (e.g., anhedonia or mood instability). To investigate these processes, I integrate fMRI, pupillometry, behavioral experiments, time-intensive self-reports, and computational modeling.

Keywords: computational psychiatry, neuroeconomics, decision-making, reward, affect

Education

PhD in Psychology Stanford University, USA	2022-2027 (expected)
MSc by Research in Psychiatry (pass with no corrections) University of Oxford, UK	2021-2022
MSc in Clinical and Therapeutic Neuroscience (distinction) University of Oxford, UK	2020-2021
Visiting Student in Experimental Psychology University of Oxford, UK	2018-2019
BSc in Applied Psychology (first place) Nanjing University, China	2016-2020

Research Experiences

Symbiotic Project on Affective Neuroscience (SPAN) Lab, Stanford University PhD researcher working with Professor Brian Knutson	2022-present
Polrack Lab, Stanford University Lab affiliate working with Professor Russell Poldrack	2024-present
Max Planck Institute for Biological Cybernetics Visiting researcher working with Professor Peter Dayan	2025
Laboratory of Decision Dynamics, University of Oxford MSc student working with Professor Laurence Hunt and Dr. Cameron Hassall	2020-2022
Computational Psychiatry Lab, University of Oxford MSc student working with Professor Michael Browning and Dr. Chamith Halahakoon	2020-2022
Stanford Psychophysiology Lab, Stanford University Visiting researcher working with Dr. Maia ten Brink and Professor James Gross	2019
IQ/EQ lab, Nanjing University	2017-2020

Undergraduate researcher working with Professor Renlai Zhou

Publications

Working Papers

14. Mortazavi, L., Gore, F., Hernandez, M., **Yan, Y.**, Deng, W., Deisseroth, K., Knutson, B., Projections from insula to accumbens drive avoidance of negative outcomes in humans and rats
13. **Yan, Y.**, Westwater, M., Juechems, K., Hunt, L., Browning, M., Anhedonic realism: Efficient two-dimensional reward trade-offs in individuals with anhedonia
12. **Yan, Y.**, Mortazavi, L., Yang, C., Knutson, B., Hedonic Capacity and Positive Arousal Differentially Relate to Univariate and Multivariate Neural Signatures of Reward Anticipation.
11. **Yan, Y.**, Wu, C., Srirangarajan, T., Dayan, P., Knutson, B., Temporal Dynamics in the Neural Representation of Positive and Negative Affect

Peer-reviewed Articles

10. (In press at *Psychological Science*) **Yan, Y.**, Halahakoon, D. C., & Browning, M. (2026). Evaluating the ecological validity and mechanism of a generative model-based decomposition of affective variability. osf.io/preprints/psyarxiv/zafjn_v1
9. (In press at *Journal of Neuroscience*) **Yan, Y.**, Srirangarajan, T., Bailenson, J., Knutson, B., (2026). Pupil size dynamics predict momentary changes in self-reported arousal.
8. Xu, N.#, **Yan, Y#.**, Saunders, K. E. A., Geddes, J. R., & Browning, M. (2025). Effect of lithium on circadian activity level and flexibility in patients with bipolar disorder: Results from the Oxford Lithium Trial. *eBioMedicine*, 115, 105676. <https://doi.org/10.1016/j.ebiom.2025.105676>
7. **Yan, Y.**, Hunt, L. T., & Hassall, C. D. (2024). Reward positivity affects temporal interval production in a continuous timing task. *Psychophysiology*. <https://doi.org/10.1111/psyp.14589>
6. Hassall, C. D., **Yan, Y.**, & Hunt, L. T. (2023). The Neural Correlates of Continuous Feedback Processing. *Psychophysiology*. <https://doi.org/10.1111/psyp.14399>
5. Zhang, J.-X., ten Brink, M., **Yan, Y.**, Goldstein-Piekarski, A., Krause, A. J., Manber, R., Kreibig, S., & Gross, J. J. (2023). Daytime affect and sleep EEG activity: A data-driven exploration. *Journal of Sleep Research*, e13916. <https://doi.org/10.1111/jsr.13916>
4. Hamilton, E. M., Rassam, W., **Yan, Y.**, Singh, A., Ng, S. Y. A., Zhang, J., ... & Chen, Z. (2023). Correlates of chronic hepatitis B virus infection in the general adult population of China: Systematic review and meta-analysis. *Journal of Viral Hepatitis*.

3. Ten Brink, M., **Yan, Y.**, Zhang, J., Goldstein-Piekarski, A., Krause, A., Kreibig, S., ... & Gross, J. (2023). Pre-sleep affect predicts subsequent REM frontal theta in nonlinear fashion. *Cognitive, Affective, & Behavioral Neuroscience*, 1-17.
2. Liu, L.#, **Yan, Y.#**, Nazhalati, N., Kuerban, A., Li, J., & Huang, L. (2020). The effect of PM2. 5 exposure and risk perception on the mental stress of Nanjing citizens in China. *Chemosphere*, 254, 126797
1. 严严#,黄玲#,华弥之,陈思佚,杨萌,周仁来. 幼儿情绪能力训练的理论构思与实证检验 [J]. 心理学通讯. 2018 (01). **Yan, Y.#**, Huang, L.#, Hua, M., Chen, S., Yang, M., & Zhou, R., et al., (2018)., The Theoretical Conception and Empirical Test of an Emotional Competence Training Program for Preschool Children. *Psychological Communications.*, 2018, 1(1): 5-15, DOI: 10.12100/j.issn.2096-5494.218001.

Joint first authors.

Other publications

(Chinese Mandarin translation work) Zhou, R., **Yan, Y.**, et al., *The Feeling of What Happens*, by Antonio Damasio 周仁来, & 严严. (2022, August). 当感受涌现时 [湛庐文化]. 中国纺织出版社.

Presentations

Talks

4. **Yan., Y.**, Hunt, L., Browning, M., (June 2025) Anhedonic realism: Efficient two-dimensional reward trade-offs in individuals with anticipatory anhedonia, selected talk at the 12th Symposium on The Biology of Decision Making 2025, Lyon, France
3. **Yan., Y.**, Hunt, L., Browning, M., (June 2025) Two-dimensional Reward Evaluation and Its Link to Anticipatory Anhedonia, poster spotlight at the 2025 Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM), Dublin, Ireland
2. Hassall, C., **Yan, Y.**, Hunt, L., (October 2024), Regression-based EEG analysis approaches for rapid and continuous cognition, flash talk in symposium: What is the catch with new analyses methods in psychophysiology? Society of Psychophysiological Research annual conference 2024; Prague, Czech Republic
1. **Yan, Y.**, Wu, C., Srirangarajan T., Knutson, B.,. (June 2024), Pupillary dilation predicts self-reported arousal during monetary incentive anticipation and outcomes, flash talk presented at the Bay Area Affective Science conference; Berkeley, United States

Poster Presentations

12. **Yan, Y.**, Wu, X., Srirangarajan, T., Deng, N., Dayan, P., Knutson, B., (June 2026) Distinct spatiotemporal neural signatures for positive aroused and negative aroused affect, Poster to be presented at the the Organization for Human Brain Mapping (OHBM) conference; Bordeaux, France
11. **Yan, Y.**, Wu, X., Srirangarajan, T., Deng, N., Dayan, P., Knutson, B., (October 2025) Temporal dynamics of the neural representation of positive and negative arousal, Poster presented at the Society for Neuroeconomics (SNE) conference, Boston, USA
10. **Yan., Y.**, Yang, C., Mortazavi, L., Knutson, B., (July 2025) Hedonic Capacity and Positive Arousal Differentially Relate to Univariate and Multivariate Neural Signatures of Reward Anticipation, Poster presented at the 2025 Computational Psychiatry Conference (CPC), Tübingen, Germany
9. **Yan., Y.**, Hunt, L., Browning, M., (June 2025) Two-dimensional Reward Evaluation and Its Link to Anticipatory Anhedonia, Poster presented at the 2025 Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM), Dublin, Ireland
8. **Yan, Y.**, Wu, X., Srirangarajan, T., Yang, L., Zhang, E., Knutson, B., (October 2024) Inferring affect from neural activity elicited by monetary incentives, Poster presented at the Society for Neuroeconomics (SNE) conference, Cascais, Portugal
7. **Yan, Y.**, Mortazavi, L., Knutson, B., (July 2024), Towards a Neurobehavioral Phenotype of Human Hedonic Capacity, Poster presented at the Organization for Human Brain Mapping (OHBM) conference; Seoul, Korea
6. **Yan, Y.**, Srirangarajan, T., Knutson, B., (October 2023), Pupillary dilation predicts self-reported arousal during monetary incentive anticipation and outcomes, Poster presented at the Society for Neuroeconomics (SNE) conference; Vancouver, Canada
5. **Yan, Y.**, Westwater, M., Hunt, L., Browning, M., (August 2023), Two-dimensional Reward Evaluation and Its Relevance to Anhedonia, Poster presented at the Conference on Cognitive Computational Neuroscience (CCN); Oxford, UK
4. **Yan, Y.**, Hunt, L., Hassall, C., (July 2022) Feedback-related neural activity observed in a drumming task predicts subsequent timing performance., Poster presented at the Federation of European Neuroscience Societies Forum (FENS); Paris, France
3. Ten Brink, M., **Yan, Y.**, Zhang, J., Manber, R., Gross, J., (June 2022)., Pre-Sleep Affect and REM Frontal Theta in a Naturalistic Context, Poster presented at the APSS Sleep Conference; Charlotte, NC, United States
2. **Yan, Y.**, Hunt, L., Hassall, C., (April 2022) The differential effect of prediction error and information on timing in a drumming task., Poster presented at the 7th Computational Properties of Prefrontal Cortex Workshop (CPPC); Oxford, United Kingdom

1. **Yan, Y.**, Liu, L., Nazhalati, N., Kuerban, A., Huang, L., (October 2019). The Effect of Dust Haze Exposure and Risk Perception on the Mental Stress of Nanjing Citizens, Poster presented at the 11th Symposium of Chinese Psychologists (第十一届两岸心理学家学术研讨会); Nanjing, China

Fellowships, Awards, and Honors

2025	Stanford University Ric Weiland Graduate Fellowship (\$4000; one per department)
2025	Stanford University Pathways to Neurosciences Fellowship (\$4000)
2025	Stanford University Graduate Research and Internship Program in Germany (\$9000)
2023	Stanford University Mind, Brain, Computation and Technology (MBCT) Center Fellowship (\$1500)
2022	Rhodes House Warden's Discretionary Fund (conference travel grant) (£330)
2020	Honorable Graduate of Jiangsu Province, China
2020	Outstanding Undergraduate Dissertation of Jiangsu Province for the senior dissertation ' <i>Frontal Alpha Asymmetry and Psychopathology: a Data-driven Approach.</i> '
2020	Rhodes Scholarship for China (1 out of 4 nationwide; £94,640)
2018	Nanjing University 'Excellence Program' (financial support for international exchange) (¥10,000)
2018	First Award, the People's Scholarship of China (¥3,000)

Teaching

Teaching Assistantships

2026	PSYCH 45: Learning and Memory (Guest lecture on Emotion and Memory)
2023, 2025	PSYCH 10: Introduction to Statistical Methods: Precalculus
2024	PSYCH 90: Introduction to Clinical Psychology: A Neuroscience Perspective
2024	PSYCH 204A: Human Neuroimaging Methods (Guest lecture on TMS treatment for depression)

Other teaching experiences

2023, 2024	Stanford Psychology PhD Stats Bootcamp
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Mentorship

Nupur Gupta	Visiting summer research intern → Undergraduate at UCLA
Helen Scarborough	Summer research intern → Undergraduate at Stanford University
Natalie Carie Wang	Honors Thesis Student → Amae Health

Dana Becker	High school research assistant → Undergraduate at Boston University
Clara Ladan Yang	Visiting summer research intern → Undergraduate at Smith College
Esther Luoyuan Zhang	Undergraduate research assistant at UCL → MSc in the MACSS program, the University of Chicago

Community Service

2025	Student representative for the Affective Area faculty search
2024-present	Departmental Tea-Time Committee
2023-2024	Organizer for the Affective Seminar
2023, 2024	Speaker and mentor for the <i>Pathway to PhD</i> program

Ad hoc Co-Reviewer

Biological Psychiatry
Neuroimage
Biological Psychiatry
Human Brain Mapping
PNAS
Schizophrenia Bulletin

Skills

Programming: Python (including Nilearn), MATLAB (including SPM and EEGLab), R, shell scripts

Research Methods: fMRI (data collection and analysis), Pupillometry (data collection and analysis), EEG (data collection and analysis), online experiments, Experience Sampling Methods, computational modelling (mainly RL models)

Language: Chinese Mandarin (native), English (full professional proficiency), Cantonese (limited working proficiency), German (A2), Spanish (A2)

Referees

Professor Brian Knutson
Department of Psychology, Stanford University, knutson@stanford.edu

Professor Russell Poldrack
Department of Psychology, Stanford University, russpold@stanford.edu

Professor Michael Browning
Department of Psychiatry, University of Oxford, michael.browning@psych.ox.ac.uk